AUD-YOLO: A Lightweight Object Detection Algorithm Model Incorporating Dynamic Upsampling for Remote Sensing Images

Xite Wang, Yinglong Wang, Mei Bai, and Qian Ma

***Abstract*—In the field of remote sensing image processing, remote sensing image object detection is a crucial undertaking. However, the existing object detection algorithms have a considerable number of model parameters, which results in a slow detection speed that is not conducive to application deployment and real-time detection. In addition, due to the background complexity of remote sensing images and the large number of small target objects, the performance of existing algorithmic models applied directly with remote sensing images is unsatisfactory. To address the aforementioned issues, this article proposes an efficient and lightweight architecture based on YOLOv8: AUD-YOLO. In particular, a convolution module, GEConv, based on improved efficient multiscale attention (EMA), is proposed to replace the original ordinary convolution at the end of the backbone part. This can enhance the accuracy of object detection in remote sensing images while maintaining a lightweight operation. The utilization of dynamic upsampling operators serves to perform upsampling operations with greater accuracy and to more effectively extract image features pertaining to small target objects. Finally, the issue of sample imbalance is addressed by utilizing Wise-iou (Wiou) as the position coordinate loss function of the neural network. In the experiments, we validate the effectiveness and robustness of the method using the public remote sensing datasets NWPU VHR-10 and DIOR. The experimental results demonstrate that the AUD-YOLO model proposed in this article achieves mAP values of 91.1% and 82.7%, which are 1.6% and 1.5% higher than the mAP indexes of the baseline model, and the number of parameters is reduced by 4.7% compared with the original model. We comprehensively consider the number of model parameters and model detection accuracy to make the model more adept at detecting remote sensing images.**

***Index Terms*—Dynamic upsampling, object detection, remote sensing image, You Only Look Once (YOLO).**

I. Introduction

OBJECT detection is a crucial field of computer vision [1], with applications in a range of domains, including autonomous driving, face recognition, and image analysis [2], [3], [4], [5], [6], [7]. The advent of advanced information technology and remote sensing technology has resulted in an increasing number of remote sensing images, which are used to describe a vast array of objects on the Earth's surface, including aircrafts, ships, and bridges. Object detection in remote sensing images, such as the detection of ships, aircraft, or other objects, is important in both military and civilian applications. However, since remote sensing images differ from natural scene images, which often have a large field of view, complex backgrounds, and many small objects, the performance of existing algorithmic models directly applied to remote sensing images is unsatisfactory. Furthermore, in real-world applications, it is necessary to embed the model of the detection algorithm into flying devices such as UAVs with limited storage capacity, which places higher demands on the lightweight operation of the algorithm. Therefore, it is necessary to design an algorithmic model dedicated to object detection with remote sensing images. The object detection algorithms of traditional remote sensing images have limited generalization capabilities and are unable to perform accurate detection work in different scenes. In recent years, there has been a surge in the development of deep learning algorithms [8], which have led to the emergence of corresponding object detection algorithms. The current common object detection algorithms can be broadly categorized into single-stage and two-stage algorithms. First, the single-stage algorithm comprises a single network model that performs both object classification and position estimation. The generation of candidate frames is performed directly on the input image, and then a classifier is used to classify and regress the generated candidate frames. Since the algorithm requires only one stage to complete the detection task, it has a low computational cost and a fast detection speed. However, when detecting image information containing small object objects, the algorithm is prone to high false detection and omission rates due to the large number of candidate frames generated and the small size of the objects. Typical single-stage object detection algorithms include Single-Shot Multi-Box Detector (SSD) [9] and You Only Look Once (YOLO) [10]. Second, two-stage object detection algorithms comprise two independent network models: the region proposal network (RPN) for generating candidate boxes and the classification and regression network for the generated detection boxes. The algorithm model initially employs the RPN network to generate candidate frames

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for the input image, after which the candidate frames are classified and localized. Typical two-stage object detection algorithms include Faster-RCNN [11] and Mask-RCNN [12]. Two-stage detection algorithms employ two independent networks in tandem to perform two computations, resulting in more accurate detection in complex images. However, the two-stage computation necessitates a longer detection time, which is less desirable in certain contexts.

In summary, object detection algorithms have been the subject of extensive research and development over an extended period. Although convolution-based object detection algorithms significantly enhance the efficiency and effectiveness of detection in comparison to traditional algorithms, their algorithmic models still exhibit shortcomings, including a weaker generalization ability, larger models due to the excessive amount of parameters that are not suitable for deployment on storage-limited devices such as UAV terminals, and so on. This article presents a novel single-stage detection algorithm, AUD-YOLO. The primary contribution of this article is described below.

1) A novel convolution module, GEConv, is proposed as a replacement for the original ordinary convolution at the end of the backbone part. GEConv is based on improved efficient multiscale attention (EMA) [18] combined with GSConv [17] and can be operated in a lightweight manner while improving the object detection accuracy of remote sensing images.

2) A lightweight and improved dynamic upsampling operator [20] is employed to enhance the upsampling accuracy and effectively extract the image features of small objects in remote sensing images. In order to improve the accuracy of the detection speed of the model as much as possible, the original upsampling method of the grid sample in the sampling operator was improved.

3) The Wise-iou (Wiou) function [19] is employed as the positional coordinate loss function of the neural network, thereby effectively addressing sample imbalance and accelerating the convergence speed of the model.

Based on YOLOv8, the model network architecture is reconstructed, experimented, and verified on several large datasets. This process greatly reduces the number of parameters of the object detection algorithm model while improving the detection accuracy. A compromise between the number of parameters and the performance effect is obtained.

The remainder of this article will be summarized as follows. In Section II, the related literature and theories will be recounted. In Section III, our proposed solution will be discussed in detail. In Section IV, the corresponding setup, initialization conditions, results, and analysis of the experiments will be shown. Finally, in Section V, the article will be summarized.

II. Related Work

A. Object Detection of Remote Sensing Image

The field of object detection in remote sensing images is of great importance in both military and civilian applications. Despite there have been many successful approaches in this field, there remain significant challenges, including the need to handle a vast number of parameters and the difficulty of detecting small objects. Researchers have made numerous contributions to address these challenges. Liu et al. [13] proposed an object detection model for remote sensing images. They replaced the original downsampling layer with the proposed lightweight SPD-Conv module, which enhances the network's ability to learn feature representations. A pyramid spatial attention mechanism was proposed to obtain weight information for different spatial locations through a hierarchical pooling operation. Ma et al. [15] improved Faster R-CNN and proposed a method that can recognize many small and medium-sized objects in large-sized images. The core of the method employs the HRNet feature extraction network to enhance small objects in the input image for detection. Zhu and Miao [16] proposed an efficient remote sensing image detection network, SCNet, which introduces selective feature enhancement blocks (SFEBs) while using a context-based transformer block at the end of the backbone. Wang et al. [14] proposed an attention-based mechanism and an efficient and lightweight YOLO detection model, AMEA-YOLO, which employs FasterNet to accelerate the training of the model. The enhanced depth second order channel attention module (EnhancedSOCA) is employed to enhance the image resolution. In addition, scholars have also conducted many explorations in lightweight work in remote sensing image object detection. For example, Chen et al. [29] proposed EFS-YOLO. It simplifies the backbone network of YOLO, adopts efficient convolution modules and reduces the number of parameters and computation, lowers the hardware deployment threshold, and solves the problems of multiple small targets and complex backgrounds in this scenario.

B. Attention Mechanism

At present, the attention mechanism is a widely used approach in the field of object detection. It involves assigning different weights to the input feature map, which enables the neural network to focus on the most important features, thereby significantly improving the detection accuracy. Hu et al. [21] proposed the squeeze and excite network (SENet) to model the interchannel links of CNN networks and also regulate the feature response values of each channel. However, SENet ignores the attentional information of the space in the input feature map. Li et al. [22] proposed the spatial group-wise enhancement (SGE) module as a means of enhancing semantic information while simultaneously suppressing the effect of noise. However, the approach in question fails to consider the channel attention information present in the input image and ignores the channel attention information of the input image. Bottleneck attention module (BAM) and convolutional block attention module (CBAM) were proposed in [23] and [24], respectively. Numerous experiments have shown that networks combining spatial and channel attention work better than using one of them. BAM combines channel attention and spatial attention in parallel and successively uses multiple dilation convolutions to expand the sensory field in the spatial dimension. In contrast, CBAM combines channel and spatial attention in tandem and employs both maximal pooling and average pooling to enrich the input's

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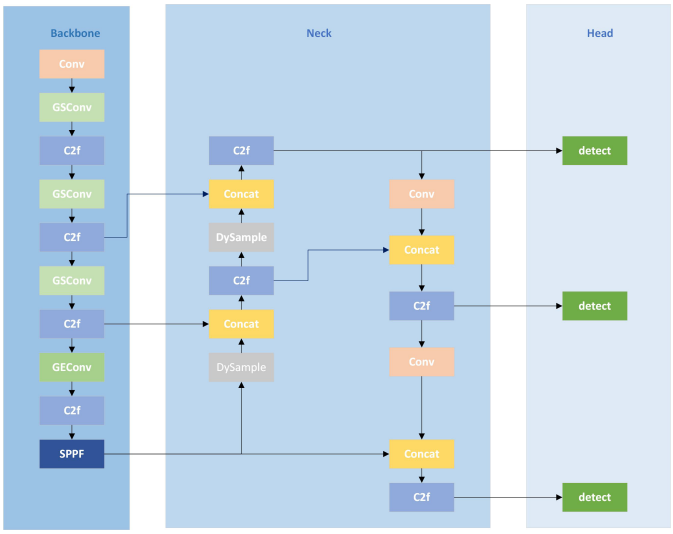


Fig. 1. AUD-YOLO model network structure.

perceptual field in different dimensions. FcaNet [25] employs different discrete cosine transforms (DCTs) to initially capture the individual intrinsic information of the channel groupings, and subsequently utilizes a fully connected layer with a bottleneck structure to facilitate the interaction of the global information. Finally, it is possible to improve the detection of the algorithmic model by using a reasonable attention mechanism. Feng et al. [30] published U3UNet: an accurate and reliable segmentation model for forest fire monitoring based on UAV vision. It enhances the ability to capture targets of different sizes through third-order feature fusion, and incorporates attention mechanisms to solve problems such as terrain background interweaving and blurred flame edges.

III. Proposed Method

A. AUD-YOLO Model Introduction

The YOLO series of models is one of the most widely used object detection algorithm models in recent years. It comprises a backbone part for coarse feature extraction, a neck for fusing image features, and a detector head for obtaining the detection results. AUD-YOLO is a model based on YOLOv8 designed to recognize small objects in remote sensing imagery. This model optimizes the YOLOv8 model to the extent that its parameters are greatly reduced while allowing for more accurate localization and extraction of relevant features for objects in remote sensing images. Briefly, we combine the advantages of GSConv and our proposed GEConv to replace the ordinary convolution in the backbone part, use the improved dynamic upsampling operator to replace the normal upsampling in the neck part, and finally adopt Wiou, which is more suitable for the training of remote sensing images, as a more suitable loss function for this model. The network structure is depicted in Fig. 1. The specifics of each module are elucidated in Sections III-C–III-E.

B. Model Structure Improvement Strategy

First, due to the need to recognize the edge information of the target object and the original convolutional module is more

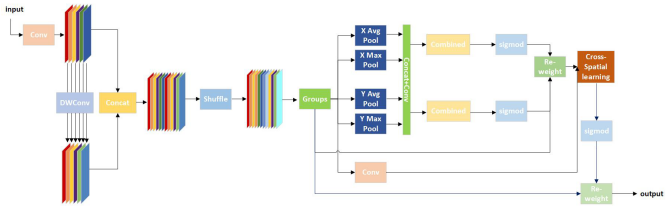


Fig. 2. GEConv network structure.

weight, the GEConv convolutional model is designed, which integrates the convolutional neural network (GSConv) and the improved attention mechanism (EMA Attention). Initially, the intention was to replace all the convolutional modules in the backbone with the GEConv module, which would have increased the accuracy. But it would have increased the number of parameters in the overall neural network. Based on weighing the advantages and disadvantages of lightweighting and precision, it was decided that only the last part of the backbone should be changed to the GEConv module. The others were simply replaced with GSConv modules. Second, the first step is to carry out the image upsampling operation in the neck. This affects the results of subsequent convolution, downsampling, and other operations. Therefore, the upsampling operation is a critical step in fusing image features in the neck. The upsampling technique in YOLOv8 is implemented using the UpSample method in the nn module of the Pytorch framework. However, these methods have inherent limitations that may result in the loss of key details of the input image. Furthermore, traditional kernel-based upsampling techniques tend to have a high parameter count and significant computational requirements. This is not conducive to achieving a lightweight network architecture. This article introduces a novel lightweight dynamic upsampling operator based on point sampling, which avoids the burden of huge computational cost while maintaining satisfactory accuracy. Eventually, the loss function is a crucial element in the model training process, influencing both the speed of model convergence and the effectiveness of training. The default loss function in YOLOv8 is Ciou. In this article, the loss function is changed to Wiou. Our experiments have demonstrated that the loss function is more suitable for object detection in remote sensing images. This is because there are many small objects in remote sensing images, and the location of the objects is more random, which greatly increases the difficulty of object detection and the number of unreasonable samples. Wiou addresses the issue of unreasonable samples by incorporating a penalty term for such samples. This approach allows for a more focused attention on these samples, leading to enhanced model training and accelerated convergence, whereas loss functions such as Ciou's are not taken into account.

C. GEConv

We propose the convolutional module GEConv. The overall result of GEConv is illustrated in Fig. 2. It is composed of our improved EMA and lightweight convolutional GSconv. First, lightweight convolution GSconv posits that standard

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convolution (SC) and depth separable convolution (DSConv) can be employed in concert. Specifically, the input image is first subjected to one SC to generate the result r_1 and reduces the number of channels by half. This result is then subjected to DSConv to obtain the result r_2. Finally, r_1 and r_2 are subjected to one Concat operation, which enables the generation of channel-enriched augmented tensors followed by a shuffle of a homogeneous mixing strategy. The EMA mechanism is a parallel attention mechanism. Instead of three parallel paths, we use five parallel paths for the extraction of the input feature map grouping attention weight descriptors. Four of the parallel paths are 1×1 branches, while the remaining fifth is a 3×3 branch. Specifically, in order to capture as many channel dependencies as possible while reducing the computational cost, in the 1×1 branch of the EMA, we add 1-D global maximum pooling based on the 1-D global average pooling of the inputs, and then integrate the pooling results in the horizontal and vertical directions with weighted averages combine. This combined average pooling and maximum pooling result facilitates the highlighting of small objects in the image for further detection. In the 3×3 branch, it utilizes a stacked 3×3 convolutional kernel to capture the multiscale feature representation of the input image. This mechanism reduces the time required to process the data and utilizes parallel processing of features at different scales, thereby improving the accuracy of detection.

The application of the attention mechanism allows more attention to be paid to special features in the image, which facilitates better feature extraction and fusion of regions of interest by the model. The utilization of GSConv can minimize the redundant information and computational cost of convolution, thereby improving the efficiency of the subsequent attention module. In addition, the EMA module is highly suitable for object detection in remote sensing images due to its ability to focus on the features of small objects, which can enhance the detection accuracy of GSConv. Therefore, we propose the GEConv module and utilize it for the convolution of object detection in remote sensing images.

D. DySample

In the object detection task, an upsampling operation is required to resize the feature map so that the model can detect objects of different sizes in the input feature map. In this article, DySample, which is a sampler in terms of point sampling and learning sampling, is introduced in this article as an upsampler for the model.

First, the sample set is generated by the sample point generator and then sampled using the grid sampling function. Second, its sample set consists of the sum of the original grid positions G and the generated offsets O. It has two versions, static range factor and dynamic range factor, and the difference lies in the way the offsets are generated; the former is generated using a linear layer, while the latter first generates the range factor and then uses it to adjust the offsets. The upsampling method used for the grid sampling function in this adopter is bilinear interpolation, and the method for handling the boundary values of a filled image is border. This method is applicable to natural images, but there may

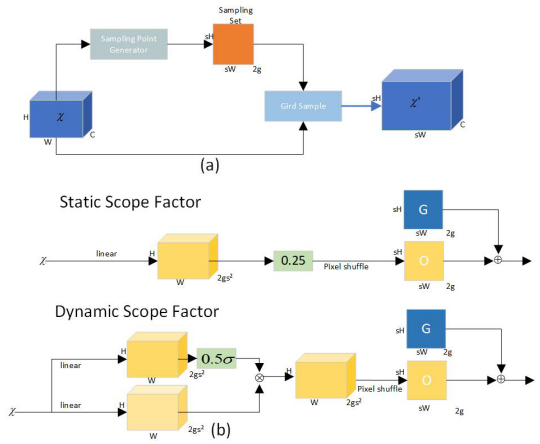


Fig. 3. DySample network structure. (a) Static scope factor. (b) Dynamic scope factor.

be many small targets located at the image boundaries in remote sensing images, and the direct use of this upsampling method will lose many details, leading to poor detection. So in this article, we modify the grid sample. The basic idea of our approach is to combine the advantages of bilinear interpolation and nearest neighbor interpolation. There are two common upsampling methods: bilinear interpolation and nearest neighbor interpolation: bilinear interpolation estimates the value of a target pixel from the values of four neighboring pixels, this method produces a relatively smooth image but may cause a blurring effect in images with sharp edges or rich details; the proximity interpolation method, on the other hand, interpolates the new pixel by setting its value directly to the value of the nearest known pixel, which is simple to implement but may lead to significant blocking effects and insufficiently smooth images. Specifically, first, we set the method of bilinear interpolation to fill the boundary information as reflection, which makes the smooth transition of image edges more natural by filling the edge pixels of the reflection image, avoiding the appearance of stretching and artifacts mixing. And when encountering sharp edges of the image can reduce the edge blurring problem to in certain extent. Second, we set the nearest neighbor interpolation to handle the boundary information filling method to border. This strategy is a simple boundary processing strategy that repetitively fills the pixel values of edges. This strategy works well with sharp image edges and reduces the risk of edge "blur diffusion" problems, but produces stepped artifacts. Therefore, the weighting ratio is used to control the two strategies (the sampling result is $a \times \text{Nearest} + (1 - a) \times \text{Bilinear}$). Balance can be found between detail retention and smooth transitions, which can have the effect of reducing artifacts, avoiding blurring problems, and keeping the edges sharp while smoothing out the center. In summary, this approach combines the advantages of both interpolation and boundary processing methods, resulting in a better balance of boundaries and details.

The network structure of DySample is shown in Fig. 3. The module [in Fig. 3(a)] is dynamic upsampling; the module [in Fig. 3(b)] shows the sampling point generator in DySample. The input image feature, upsampled feature, generated offset,

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and original grid are denoted by X, X', O, and G, respectively. σ is the sigmoid function. The symbol sH represents the sampled height, sW represents the sampled width, $2gs^2$ is the number of channels after the feature graph passes through the linear layer.

E. Loss Function Improvement

A reasonable loss function is beneficial to the optimized training of the model. The default loss function used in the Baseline model is Ciou. The Diou takes into account the distance between the predicted box and the real box, but does not consider the aspect ratio of the bbox regression triad in the calculation. This is solved by Ciou. A summary of Ciou's expression is as follows:

$$L_{\mathrm{CIOU}} = 1 - \mathrm{IOU} + \frac{\rho^2\left(b, b^{\mathrm{gt}}\right)}{(w^c)^2 + (h^c)^2} + \alpha v. \tag{1}$$

Among that

$$\alpha = \frac{v}{(1 - \mathrm{IOU}) + v} \tag{2}$$

$$v = \frac{4}{\pi^2}\left(\arctan\frac{w^{\mathrm{gt}}}{h^{\mathrm{gt}}} - \arctan\frac{w}{h}\right)^2 \tag{3}$$

where assume that w and h denote the width and height of the prediction box, respectively, while w^{gt} and h^{gt} represent the width and height of the real box. The centroid of the boundary between the prediction box and the real box is denoted by b and b^{gt}, while ρ denotes the Euclidean distance between b and b^{gt}. Finally, w^c and h^c represent the width and height of the smallest external rectangle of the prediction box and the real box, respectively, while IOU represents the ratio of intersection over union.

The majority of contemporary mainstream loss functions assume that the input images in the dataset are of high quality and prioritize the regression of the bounding boxes of these samples. However, when the dataset contains a large number of remote sensing images, it frequently includes a considerable number of low-quality samples. For these low-quality samples, if the same method of calculating the loss is also sampled, it is likely to result in geometric factors such as aspect ratio amplifying the negative gradient of the low-quality samples, which in turn negatively affects the detection performance of the model. Therefore, in this article, Wiou is used as the loss function of the neural network model instead of the original Ciou. This approach introduces a dynamic nonmonotonic focusing mechanism that differs from other loss functions, such as Ciou and Eiou. This allows for a greater focus on common quality samples during training, and evaluates the quality of the anchor frames by using anomalies rather than IoUs. In addition, a sensible gradient gain assignment strategy is provided, which enhances the overall performance of the model. It is assumed that (x,y) corresponds to $(x^{\mathrm{gt}},y^{\mathrm{gt}})$ in the object frame. In this context, RWIOU represents the loss of the high-quality anchor frame. The expression for Wiou v1 is available as follows:

$$\mathcal{L}_{\mathrm{WIoUv1}} = \exp\left(\frac{\left(x - x^{\mathrm{gt}}\right)^2 - \left(y - y^{\mathrm{gt}}\right)^2}{\left((w^c)^2 + (h^c)^2\right)}\right) \cdot \mathcal{L}_{\mathrm{IoU}}. \tag{4}$$

Among that

$$\mathcal{L}_{\mathrm{IoU}} = 1 - \mathrm{IoU}. \tag{5}$$

To effectively address the issue of harmful gradients that arise due to the inclusion of low-quality samples in the dataset, Wiou v3 is generated on the foundation of Wiou v1 and the following equation:

$$L_{\mathrm{WIOU}_{V3}} = rL_{\mathrm{WIOU}_{V1}}. \tag{6}$$

Among that

$$r = \frac{\beta}{\delta\alpha^{\beta-\delta}} \tag{7}$$

$$\beta = \frac{L^*_{\mathrm{IOU}}}{\overline{L_{\mathrm{IOU}}}}. \tag{8}$$

In this context, β represents the quality of the anchor frame, r is a nonmonotonic aggregation factor that effectively reduces the harmful gradient problem due to ordinary quality samples, L^*_{IOU} is a monotonic aggregation factor, while $\overline{L_{\mathrm{IOU}}}$ is the sliding average of L_{IOU}. Finally, α and δ are hyperparameters. In this context, the value of δ leads to $r = 1$ when $\beta = \delta$, the anchor box will enjoy the highest gradient gain when its outlier degree satisfies $\beta = C$, where C is a constant value. Since the L_{IOU} changes dynamically, the quality classification criteria for the anchor box are also constantly changing, allowing WIoU v3 to determine the best gradient gain allocation strategy at any given time.

IV. Experiment and Discussion

A. Datasets

This article employs two datasets: NWPU VHR-10 and DIOR [35], [36]. NWPU VHR-10 is a commonly used dataset, released by the Northwestern Polytechnical University in 2015 and comprising 800 VHR optical remote sensing images. The dataset is divided into two categories: the dataset comprises 650 images with at least one object present in the image, and 150 images with no object present in the image. The dataset comprises a total of ten geospatial object categories. The images were divided into a training set and a test set in a ratio of 8:2. The DIOR dataset is a comprehensive and challenging benchmark for optical remote sensing images, comprising a total of 23 463 images and 1 90 288 instance objects across 20 categories. It has been developed by the Technical University of Munich and the Northwestern Polytechnical University, and includes image information under different weather and seasons. The specific number of instance objects belonging to distinct categories in the NWPU VHR-10 and DIOR datasets is presented in Figs. 4 and 5.

B. Experiment Setting

The experiments described in this article were conducted on the Windows 11 operating system using the PyTorch deep learning framework and CUDA 12.1 with the Python version 3.9.13. The primary equipment for the experiment was a PC builded by a NVIDIA GeForce RTX 3090Ti graphics card and a 12th Intel[1] Core[2] i5-12600KF. In the model training phase,

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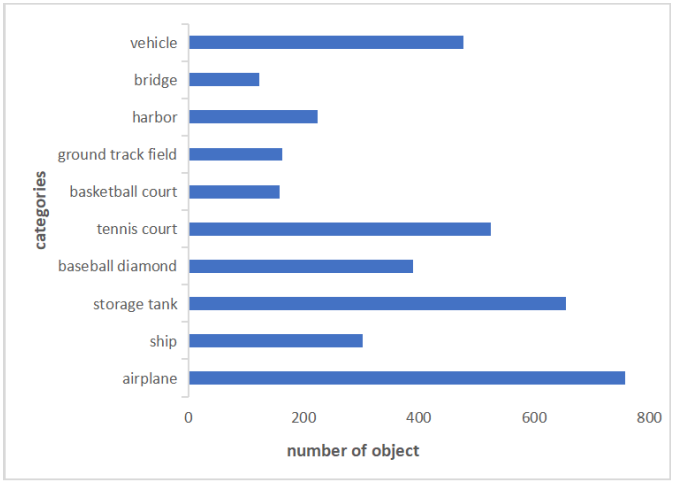


Fig. 4. Number of objects of different categories in the NWPU VHR-10.

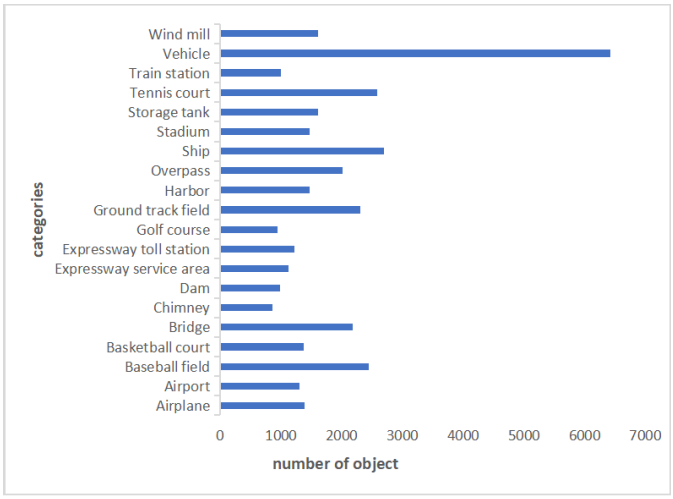


Fig. 5. Number of objects of different categories in the DIOR.

the initial learning rate was set to 0.01, the final learning rate was also set to 0.01, the momentum parameter was set to 0.937, the Adam optimizer was used for the neural network to update the parameter weights, and the IOU was set to 0.7. The batch size was set to 16, and the epoch was set to 1000, and the remaining parameters were set to their default values. The parameter a in the improved dynamic upsampling grid sampling function is 0.102.

C. Evaluation Metrics

For validating the effectiveness of the method proposed in this article, the algorithm is evaluated using precision rate, recall rate, average precision, mean of average precision, and the number of method model parameters.

Precision rate indicates the proportion of correct predictions of the algorithm model in samples that have been judged positive. It can be represented by

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}. \tag{9}$$

Recall rate indicates the proportion of samples that have been labeled positive that have been predicted correctly by the algorithm model. It can be defined as

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{10}$$

The area under the precision–recall curve and the coordinate x-axis is referred to as AP. The mean average precision (mAP) is the average of the AP values across all categories in the dataset. This is calculated as

$$\mathrm{AP} = \int_0^1 P(y)\, dy \tag{11}$$

$$\mathrm{mAP} = \frac{\sum_n^N \mathrm{AP}_n}{N}. \tag{12}$$

In the above formula, TP represents the number of true positive samples, TN represents the number of true negative samples, FP represents the number of false positive samples, and FN represents the number of false negative samples, and y is the precision–recall curve.

In addition, Params is the total number of parameters to be trained in the model training, and FLOPs is the number of computations that can be performed, which is a measure of model complexity.

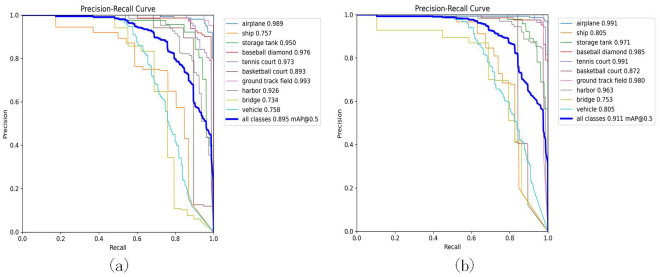


Fig. 6. PR curves of the comparison experiments on the NWPU VHR-10 dataset. (a) PR curves of the baseline model (YOLOv8n). (b) PR curves of the AUD-YOLO.

D. Results and Discussions

In this section, YOLOv6 [26], YOLOv8 [27], and YOLOv10 [28], which are newer algorithmic models with a similar number of parameters as the AUD-YOLO model, are chosen for comparison experiments. In addition, some non-YOLO series models such as ATSS [31], MobileNetV2 [32], ShuffleNet [33], and ARSD [34] are also considered in this experiment. And the more weighty algorithmic model was not chosen because the probability of detection metrics rises rapidly as the number of parameters increases. It's meaningless to compare models with several times the number of parameters of AUD-YOLO. The corresponding comparison experiments on the performance of each model on the NWPU VHR-10 dataset are presented in Table I. Besides, since the AUD-YOLO model is improved from the YOLOv8 model, this section will primarily focus on a comparative analysis of the experimental outcomes of the ADU-YOLO and YOLOv8 models.

The experimental results indicate that the number of parameters and model size of the AUD-YOLO model proposed in this article have been reduced by 4.7% in comparison to that

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TABLE I
RESULTS OF COMPARISON EXPERIMENTS ON NWPU VHR-10

Method	Params(M)	mAP50(%)	mAP50-95(%)	FLOPS(G)	size(MB)
YOLOv6	4.50	88.3	55.8	13.1	8.47
YOLOv8	3.15	89.5	55.4	8.9	6.09
YOLOv10	**2.71**	87.8	53.3	**8.4**	**5.75**
ATSS	18.96	90.95	-	89.90	-
MobileNetV2	10.29	76.90	-	71.49	-
ShuffleNet	12.10	83.00	-	82.17	-
ARSD	11.57	90.92	-	26.65	-
AUD-YOLO	3.00	**91.1**	**56.5**	8.6	5.80

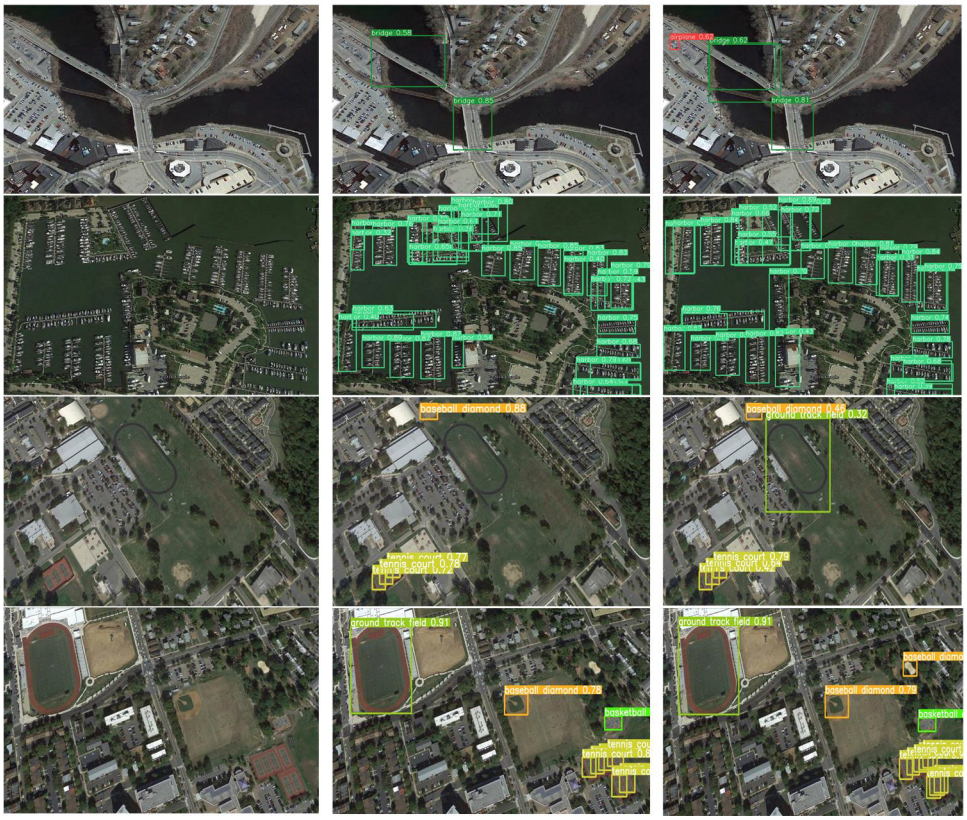


Fig. 7. Comparison of sample test results between YOLOv8n and AUD-YOLO on NWPU VHR-10 dataset. (a) Sample images from the NWPU VHR-10 dataset. (b) Detection results of YOLOv8. (c) Detection results of AUD-YOLO.

of the YOLOv8 model. Concurrently, the detection accuracy indexes of mAP50 and mAP50-95 have been increased by 1.6% and 1.1%, respectively. Fig. 6 illustrates the P-R curves of the models of the two algorithms and the information regarding the mAP50 detection results for each category, respectively. These figures serve to verify the effectiveness of the proposed method.

The detection results of a subset of samples in this dataset were also selected for comparison. The results of the YOLOv8 and AUD-YOLO algorithms are presented in Fig. 7 below. Where Fig. 7(a) is the original image, Fig. 7(b) is the detected result of YOLOv8, and Fig. 7(c) is the detected result of AUD-YOLO.

The experimental results for the YOLOv10 model indicate that the algorithm model has the smallest number of parameters, resulting in the smallest generated model volume. However, its mAP index is lower. In contrast, the YOLOv6 model has a larger number of parameters and a larger model volume, yet its overall detection accuracy is also lower than that of AUD-YOLO. In conclusion, the combined performance of the number of parameters and detection accuracy of AUD-YOLO, as proposed in this article, is the optimal solution, which validates the feasibility of the proposed method.

Finally, in order to verify the generalization ability of AUD-YOLO, comparison experiments were conducted on the DIOR dataset. The corresponding experimental results are shown in Table II. The results demonstrate that the algorithmic model exhibits the desired generalization ability.

E. Ablation Study

In order to ascertain the impact of each module proposed and utilized in this article on baseline improvement, this study was conducted to assess the performance effect of the

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TABLE II
RESULTS OF COMPARISON EXPERIMENTS ON DIOR

Method	Params(M)	mAP50(%)	mAP50-95(%)	FLOPS(G)	size(MB)
YOLOv6	4.50	71.8	49.2	13.1	8.48
YOLOv8	3.15	81.2	56.7	8.9	6.10
YOLOv10	**2.71**	80.2	56.8	**8.4**	**5.80**
ATSS	18.98	67.70	-	95.50	-
MobileNetV2	10.32	58.20	-	76.10	-
ShuffleNet	12.12	61.30	-	87.31	-
ARSD	13.10	70.10	-	41.60	-
AUD-YOLO	3.00	**82.7**	**56.9**	8.6	5.82

TABLE III
RESULTS OF ABLATION EXPERIMENTS ON NWPU VHR-10

Baseline	+Wise-iou	+GSConv	+DySample	+GEConv	mAP50(89.5)	Param(M)(3.15)
YOLOv8	✓				90.8	3.15
	✓	✓			88.3	3.11
	✓	✓	✓		89.8	3.11
	✓	✓	✓	✓	**91.1**	**3.00**
		✓			89	3.11
			✓		90.1	3.15
				✓	89.8	3.02

enhanced network model by eliminating modules such as GEConv one by one. The following Table III illustrates the specific implementation process and experimental results of the elimination experiments.

A substantial number of model parameters is associated with an increase in accuracy. Consequently, it appears to be a paradox to reduce the number of model parameters while simultaneously enhancing the accuracy of the model. The experimental results indicate that the improved mAP metrics are not consistently enhanced when the four modules are incorporated into the YOLO model in a sequential manner. First, the Wiou loss function was employed in the experiments, resulting in an improvement of 1.3% in the neural network model's mAP index. This outcome provides clear superiority of this loss function in calculating the loss of many small objects in remote sensing images. However, the loss function was unable to reduce the number of parameters, prompting the next experiment to reduce the model's complexity. The ordinary convolution operation was replaced with the lightweight GSConv in the backbone section. As illustrated in the second row of the table, the number of parameters of the model was reduced by 1% in comparison to the original algorithm. However, the accuracy was diminished by 2.5% in comparison to the model with Wiou, which indicates that this module affects the accuracy of the object detection algorithm while reducing the number of parameters. To achieve progress in parametric quantity and accuracy together, we introduced a dynamic upsampling operator in the neck for model upsampling. This operator is lightweight, thus, the parametric quantity of the model does not grow after its introduction. Furthermore, the accuracy of the detection is improved by 1.5%. However, the model is only a parametric improvement over the original model, so the model was finally improved using the GEConv module placed in the last part of the backbone. It reduces the effect of the parametric quantity and, at the same time, improves the detection accuracy due to the addition of the improved attention mechanism, as shown in the fourth row of the table. The addition of the GEConv reduces the number of parameters by 4.7% compared to the original model, while improving the mAP value by 0.3% compared to the effect of using Wiou only. From the last three rows of the table, it can be seen that in terms of overall effect, each individual effect is not as effective as the collaborative effect of the four modules. Finally, it is clear from the results of the ablation experiments that the feasibility and practicality of each design in the AUD-YOLO model proposed in this article.

V. CONCLUSION

This article presents a novel lightweight algorithmic model, AUD-YOLO, which addresses challenges of a large number of parameters, difficulty in deployment, and high leakage detection rate. A lightweight convolutional module, GEConv, based on an improved EMA mechanism, is proposed to enhance detection accuracy and reduce the number of computational parameters of the model. Furthermore, the improved DySample, a novel upsampling technique, is integrated into YOLOv8, which effectively improves the accuracy of the neck upsampling. In addition, the Wiou loss function is introduced to enhance model accuracy and accelerate model convergence speed. The feasibility and effectiveness of the algorithm proposed in this article are validated in the NWPU VHR-10 and DIOR datasets. The experimental outcomes demonstrate that the quantity of parameters is diminished from the original model parameter amount while the accuracy can be enhanced. AUD-YOLO offers a novel approach to remote sensing image detection work, yet there are disadvantages, such as a greater quantity of model layers, which can be further optimized. Future work may include the reduction of the number of detection layers of the model and the streamlining of the network architecture, while ensuring accuracy. This involves more sophisticated algorithmic techniques, combining relevant approaches from other fields or considering new architectures for further optimization of detection speed, model inference capabilities.

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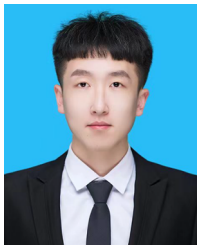
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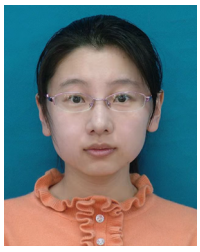
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